

COMPILER THEORY

Chapter 3: LL(1) Parsing

An LL(1) parser is a top-down parser that reads input Left to right and produces a Leftmost derivation, using 1 token of lookahead. A grammar is LL(1) if, for every pair of productions $A \rightarrow \alpha \mid \beta$, the FIRST sets of α and β are disjoint, and if ϵ is in $\text{FIRST}(\alpha)$, then $\text{FIRST}(\beta)$ and $\text{FOLLOW}(A)$ are also disjoint.

FIRST and FOLLOW Set Example

Consider the grammar $E \rightarrow T E' \mid \epsilon$, $E' \rightarrow + T E' \mid \epsilon$, $T \rightarrow F T' \mid \epsilon$, $T' \rightarrow * F T' \mid \epsilon$, $F \rightarrow (E) \mid \text{id}$. The FIRST and FOLLOW sets for each non-terminal are shown below.

Non-Terminal	FIRST Set	FOLLOW Set
E	{ (, id }	{) , \$ }
E'	{ + , epsilon }	{) , \$ }
T	{ (, id }	{ + ,) , \$ }
T'	{ * , epsilon }	{ + ,) , \$ }
F	{ (, id }	{ * , + ,) , \$ }

Chapter 4: Panic Mode Error Recovery

Panic mode recovery discards input symbols one at a time until a token in a predefined synchronizing set is found. Synchronizing tokens are usually statement or block delimiters such as semicolons or closing braces, chosen so the parser can resume without cascading further errors.